

Kazi Ahmed Asif Fuad

Ph.D. Candidate in Computer Science | Machine Learning Systems | LLMs | Hardware Acceleration

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Professional Summary

Ph.D. candidate in Computer Science at Oregon State University with research experience in machine learning systems, efficient deep learning, quantization, Kolmogorov–Arnold Networks (KANs), large language models, Generative AI, FPGA/ASIC design, and hardware acceleration. Strong background in PyTorch-based model development, GenAI/LLM workflows, quantization-aware training, post-training quantization, SystemC/C++ HLS, Vivado HLS, Catapult HLS, and ASIC design flows. Published researcher with experience across ML algorithms, accelerator design, scientific computing, simultaneous translation, and technical writing tools.

Education

Oregon State University

Corvallis, OR, USA

Ph.D. in Computer Science (AI Minor), School of EECS

Sept. 2021 – Dec. 2026 expected

- Research areas: Machine learning systems, Large language models, Joint Embedding Predictive Architecture, World models, Quantization, Sparsity, KANs, and Hardware Accelerators for ML.
- Advisor: Professor Lizhong Chen.

Erasmus Mundus Joint Master in Image Processing and Computer Vision

Hungary, Spain, France

Pázmány Péter Catholic University, Universidad Autónoma de Madrid, Université de Bordeaux

Sept. 2018 – Aug. 2020

- Master thesis: *Recognition of Sports Gestures with 3D Deep CNNs: Explanation of Networks.*
- Awarded Erasmus Mundus scholarship covering tuition and stipend; ranked 2nd among approximately 450 applicants.

American International University-Bangladesh

Dhaka, Bangladesh

M.Sc. and B.Sc. in Electrical and Electronic Engineering

2009 – 2015

- M.Sc. with Summa Cum Laude; Vice Chancellor's Best Thesis Award for both M.Sc. and B.Sc. theses.

Technical Skills

Programming: Python, C, C++, MATLAB, SystemC, Verilog, VHDL, SystemVerilog, Git
ML/DL: PyTorch, TensorFlow, ScikitLearn, OpenCV, Transformers, LLMs
Hardware/EDA: Vivado HLS, Catapult HLS, Xilinx Vivado, Cadence ASIC tools
Research Areas: Efficient ML, KANs, Accelerator design, FPGA/ASIC implementation, JEPA/World Models

Research and Professional Experience

Graduate Assistant

Sept. 2021 – Present

School of Electrical Engineering and Computer Science, Oregon State University

Corvallis, OR

- Research efficient machine learning systems with emphasis on KAN quantization, sparsity, accelerator design, LLM-based simultaneous translation and writing assistance, and JEPA-based world models.
- Developed quantization and hardware-aware methods using PyTorch, SystemC/C++ HLS, and FPGA toolchains.
- Contributed to NSF-funded research projects on ML accelerators, sparsity, and HPC interconnect optimization.

Assistant Professor, on Study Leave

Aug. 2021 – Present

Faculty of Engineering, American International University-Bangladesh

Dhaka, Bangladesh

Lecturer

Jan. 2016 – July 2021

Faculty of Engineering, American International University-Bangladesh

Dhaka, Bangladesh

- Taught and supported courses and laboratories related to digital design, FPGA design, VLSI, embedded systems, signal processing, and hardware/software co-design.

CERN Openlab Summer Student

July 2019 – Aug. 2019

European Organization for Nuclear Research, CERN

Geneva, Switzerland

- Implemented and optimized graph neural network inference on FPGA platforms for scientific computing applications using HLS-based workflows.

Research Intern, and Invited Researcher

Jan. 2020 – Aug. 2021

LaBRI, Université de Bordeaux

Bordeaux, France

- Conducted research on explainable artificial intelligence for vision tasks under the supervision of Prof. Jenny Benois-Pineau.

Selected Research Projects

QuantKAN: Quantized Kolmogorov–Arnold Networks and Hardware Acceleration 2025 – Present

Graduate Assistant, Oregon State University

- Developing a unified quantization framework for KANs across PyKAN, EfficientKAN, FastKAN, and GRAM/KAGN variants, combining QAT/PTQ, mixed-precision analysis, and branch-aware sensitivity studies.
- Exploring FPGA/HLS acceleration and hardware-aware compression for KANs using SystemC, Vivado HLS, and Catapult HLS, with emphasis on spline/basis computation, memory efficiency, vector-based hardware-aware quantization, binary quantization, and low-bit inference.

JEPA-Based World Models for Multimodal Representation Learning 2025 – Present

Graduate Assistant, Oregon State University

- Investigating Joint-Embedding Predictive Architectures for scalable representation learning across vision and text modalities.
- Exploring masked-prediction training pipelines, encoder models, and evaluation workflows using linear probes and downstream reasoning/classification tasks.

Large Language Models for Simultaneous Translation and Scientific Writing Assistance 2023 – 2025

Graduate Assistant, Oregon State University

- Designed LLM-Ref, an LLM-based reference management and citation-handling framework for technical writing.
- Contributed to Simul-LLM, a framework for exploring high-quality simultaneous translation with large language models, focusing on latency-quality trade-offs and LLM-based translation behavior.

Sparsity Exploration in Machine Learning Accelerators 2022 – Present

Graduate Assistant, Oregon State University

- Studied hardware-aware sparsity methods for neural network accelerators using HLS and ASIC design tools, including algorithm-hardware tradeoffs for efficient transformer deployment and ongoing sparsity exploration for KAN-based models.

HPC Interconnect Optimization 2022

Graduate Student, Oregon State University

- Developed edge-based topology optimization heuristics for shortcut-augmented high-performance computing interconnects to improve latency and bandwidth efficiency.

Development of TRP System for Bangladeshi Television Network 2021 – Present

Co-Project Investigator Bangladesh Satellite Company Limited

- Prototyped a top-to-bottom Television Rating Point (TRP) system for Bangladeshi television networks, covering data collection, processing, analytics, and reporting.

Selected Publications

- **K. A. A. Fuad** and L. Chen, “QuantKAN: A Unified Quantization Framework for Kolmogorov–Arnold Networks,” arXiv preprint arXiv:2511.18689, 2025.
- **K. A. A. Fuad** and L. Chen, “LLM-Ref: Enhancing Reference Handling in Technical Writing with Large Language Models,” arXiv preprint arXiv:2411.00294, 2024.
- V. Agostinelli, M. Wild, M. Raffel, **K. A. A. Fuad**, and L. Chen, “Simul-LLM: A Framework for Exploring High-Quality Simultaneous Translation with Large Language Models,” ACL, 2024.
- **K. A. A. Fuad** and L. Chen, “A Survey on Sparsity Exploration in Transformer-Based Accelerators,” *Electronics*, vol. 12, no. 10, 2299, 2023.
- **K. A. A. Fuad**, K. Zeng, and L. Chen, “Edge-Based Heuristics for Optimizing Shortcut-Augmented Topologies for HPC Interconnects,” *Electronics*, vol. 11, no. 17, 2778, 2022.
- **K. A. A. Fuad**, P.-E. Martin, R. Giot, R. Bourqui, J. Benois-Pineau, and A. Zemmari, “Features Understanding in 3D CNNs for Actions Recognition in Video,” IPTA, 2020.
- **K. A. A. Fuad** and S. M. Rizvi, “Varying Sample-Width to Realize Area-Efficient FPGA Realization of Sobel–Feldman Edge Detector,” *AIUB Journal of Science and Engineering*, vol. 14, no. 1, 2015.
- **K. A. A. Fuad** and S. M. Rizvi, “Hardware/Software Co-Simulation of Canny Edge Detection Algorithm,” *International Journal of Computer Applications*, vol. 122, no. 19, pp. 7–12, 2015.

Honors, Awards, and Service

- Oregon State University Edith McDougall Scholarship/Fellowship, Provost Scholarship, and EECS Progression Scholarship.
- Erasmus Mundus Scholarship for Image Processing and Computer Vision, covering full tuition and stipend.
- Vice Chancellor’s Best Thesis Award for both M.Sc. and B.Sc. thesis projects at AIUB.
- Reviewer for AIUB Journal of Science and Engineering and ICREST; Judge for IEEE Student Awards Program.